

MECHANICAL GENERAL NOTES

1. DRAWINGS AND SPECIFICATIONS ARE COMPLIMENTARY. WHATEVER IS CALLED FOR IN EITHER IS BINDING AS THOUGH CALLED FOR IN BOTH.
2. ALL WORK SHALL CONFORM WITH ALL APPLICABLE BUILDINGS CODES, FIRE CODES, AND ALL AUTHORITIES HAVING JURISDICTION.
3. THE EQUIPMENT SPECIFIED ON THE DRAWINGS HAVE BEEN SELECTED AS THE BASIS OF DESIGN. THE USE OF REVIEWED OR SPECIFIED EQUALS SHALL BE COORDINATED BY THE CONTRACTOR FOR SPACE REQUIREMENTS, EQUIPMENT DIMENSIONS, AND PERFORMANCE.
4. DRAWINGS ARE DIAGRAMMATIC AND SHOW THE GENERAL DESIGN INTENT, ARRANGEMENT, AND GENERAL EXTENT OF SYSTEMS. DO NOT SCALE DRAWINGS NOR USE AS SHOP DRAWINGS. WHERE ALTERNATIVE ROUTING, OFFSETS, AND TRANSITIONS ARE REQUIRED FOR FIELD COORDINATION OF ALL OTHER TRADES, THIS CONTRACTOR SHALL FIELD COORDINATE WITH ALL OTHER TRADES, AND SHALL MAKE CHANGES WITHOUT ADDITIONAL COSTS.
5. CONTRACTOR SHALL CLOSELY COORDINATE NEW MECHANICAL WORK WITH ALL NEW AND EXISTING MECHANICAL, PLUMBING, ELECTRICAL, FIRE PROTECTION, ARCHITECTURAL, AND STRUCTURAL MEMBERS. RELOCATE EXISTING MECHANICAL, PLUMBING AND FIRE PROTECTION WORK AS REQUIRED TO ACCOMMODATE ALL NEW WORK (ARCHITECTURAL, STRUCTURAL, MECHANICAL, ELECTRICAL, PLUMBING, FIRE PROTECTION, FIRE ALARM, LOW VOLTAGE, AV, ETC.
6. ALL EQUIPMENT SHALL BE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATION UNLESS SPECIFICALLY DIRECTED OTHERWISE.
7. COORDINATE ALL DIFFUSER AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, FIRE PROTECTION AND ELECTRICAL DRAWINGS.
8. INSTALL CONDENSATE DRAINS FOR ALL COOLING COILS WITH TRAP DEPTH EQUAL TO 1.5 TIMES THE UNIT'S TOTAL STATIC PRESSURE. DISCHARGE CONDENSATE TO FLOOR SINK/FLOOR DRAIN UNLESS NOTED OTHERWISE.
9. ALL ROOF WORK SHALL BE PER THE ROOFING MANUFACTURE'S INSTALLATION INSTRUCTIONS TO MAINTAIN THE EXISTING ROOF WARRANTY.
10. CONTRACTOR SHALL BE RESPONSIBLE FOR LOCATION AND INSTALLING SLEEVES, INSERTS AND SUPPORTS AS REQUIRED FOR THIS SCOPE OF WORK AND/OR CORE DRILL REQUIREMENTS. COORDINATE WITH GENERAL CONTRACTOR AND STRUCTURAL ENGINEER AS REQUIRED.
11. CONTRACTOR SHALL FIELD VERIFY ALL MECHANICAL ITEMS PRIOR TO SUBMITTING A BID.
12. PROVIDE ACCESS PANELS IN HARD CEILINGS FOR ACCESS TO ALL MECHANICAL EQUIPMENT, FIRE DAMPERS, FIRE/SMOKE DAMPERS, ISOLATION VALVES, ETC. THIS SHALL INCLUDE ALL NEW MECHANICAL ITEMS REQUIRING ACCESS.
13. PROVIDE REDLINE MARKUPS OF ANY FIELD CHANGES OR MODIFICATIONS ON THE CONSTRUCTION DOCUMENTS. REDLINE DRAWINGS SHALL BE REQUIRED WHETHER COORDINATION DRAWINGS ARE REQUIRED OR NOT.
14. THE MECHANICAL DIAGRAMS SHALL BE INCORPORATED INTO THE ASSOCIATED WORK AND PROVIDE GENERAL GUIDANCE AS TO THE INSTALLATION INTENT WHETHER REFERENCED TO OR NOT. IT IS THE CONTRACTOR'S RESPONSIBILITY TO PROVIDE ALL NECESSARY COMPONENTS FOR A COMPLETE INSTALLATION, AND INSURE THAT ALL INSTALLATIONS ARE IN ACCORDANCE WITH THE EQUIPMENT'S MANUFACTURER'S WRITTEN INSTALLATION INSTRUCTIONS.
15. COORDINATE AND VERIFY ACTUAL APPROVED EQUIPMENT DIMENSIONS PRIOR TO POURING CONCRETE EQUIPMENT PADS.

MECHANICAL LEGEND

ALL SYMBOLS IN LEGEND MAY NOT BE USED ON THIS PROJECT.					
ABBREVIATIONS					
AD	ACCESS DOOR	FLA	FULL LOAD AMPS	P	PUMP
AFF	ABOVE FINISH FLOOR	FPM	FEET PER MINUTE	PCF	POUND PER CUBIC FOOT
AFMS	AIR FLOW MEASURING STATION	FPS	FEET PER SECOND	PD	PRESSURE DROP
AFUE	ANNUAL FUEL UTILIZATION	FT	FEET	PH	PHASE
	EFFICIENCY	GAL	GALLONS	PPM	PARTS PER MILLION
AHU (AH)	AIR HANDLING UNIT	GPH	GALLONS PER HOUR	PRV	PRESSURE REDUCING VALVE
AP	ACCESS PANEL	GPM	GALLONS PER MINUTE	PSI	POUND PER SQUARE INCH
BAS	BUILDING AUTOMATION SYSTEM	HD	HEAD	PSIA	POUND PER SQUARE INCH ABS.
BOD	BOTTOM OF DUCT	HP	HORSE POWER	PSIG	POUND PER SQUARE INCH GAUGE
BOP	BOTTOM OF PIPE	HR	HOUR	RA	RETURN AIR
BHP	BRAKE HORSE POWER	HZ	HERTZ	RH	RELATIVE HUMIDITY
BMS	BUILDING MANAGEMENT SYSTEM	IN	INCH	RPM	REVOLUTIONS PER MINUTE
BTU	BRITISH THERMAL UNIT	I.E	INVERT ELEVATION	RTU	ROOFTOP UNIT
CFH	CUBIC FEET PER HOUR	KW	KILOWATT	SEER	SEASONAL ENERGY EFFICIENCY
CFM	CUBIC FEET PER MINUTE	KWH	KILOWATT-HOUR	SA	SUPPLY AIR
CO	CARBON MONOXIDE	LAT	LEAVING AIR TEMPERATURE	SP	STATIC PRESSURE
CO2	CARBON DIOXIDE	LBS	POUNDS	SQ FT	SQUARE FEET
COP	COEFFICIENT OF PERFORMANCE	LF	LINEAR FEET	TAB	TESTING AND BALANCING
CU	CONDENSING UNIT	LWT	LEAVING WATER TEMPERATURE	TDH	TOTAL DEVELOPED HEAD
CV	CONSTANT VOLUME	MAT	MIXED AIR TEMPERATURE	TEL	TOTAL EQUIVALENT LENGTH
CWP	CONDENSER WATER PUMP	MBH	1000 BRITISH THERMAL UNITS PER HOUR	TSP	TOTAL STATIC PRESSURE
DB	DRY BULB TEMPERATURE			TYP	TYPICAL
dB	DECIBEL	MCA	MINIMUM CIRCUIT AMPS	UNO	UNLESS NOTED OTHERWISE
DDC	DIRECT DIGITAL CONTROL	MERV	MINIMUM EFFICIENCY REPORTING	UV	ULTRA VIOLET
DN	DOWN		VALUE	V	VOLT
DP	DIFFERENTIAL PRESSURE	MOCP	MAXIMUM OVER CURRENT	VAV	VARIABLE AIR VOLUME
DS	DUCT SUMP		PROTECTION	VD	VOLUME DAMPER (MANUAL)
DX	DIRECT EXPANSION	NA	NOT APPLICABLE	VFD	VARIABLE FREQUENCY DRIVE
(E)	EXISTING	NC	NORMALLY CLOSED	VTR	VENT THRU ROOF
EA	EXHAUST AIR	NFPA	NATIONAL FIRE PROTECTION	WB	WET BULB TEMPERATURE
EAT	ENTERING AIR TEMPERATURE	ASSOCIATION		WC	WATER COLUMN
EER	ENERGY EFFICIENCY RATIO	NIC	NOT IN CONTRACT	WPD	WATER PRESSURE DROP
EFF	EFFICIENCY	NO	NORMALLY OPEN	(N)	NEW
ESP	EXTERNAL STATIC PRESSURE	NPSHA	NET POSITIVE SUCTION HEAD	(E)	EXISTING
ET	EXPANSION TANK	AVAILABLE		(F)	FUTURE
EWT	ENTERING WATER TEMPERATURE	NPSHR	NET POSITIVE SUCTION HEAD	(REL)	RELOCATED
	DEGREES FAHRENHEIT	REQUIRED			
FCU (FC)	FAN COIL UNIT	OA	OUTSIDE AIR		

SYMBOLS AND DESCRIPTIONS

	EQUIPMENT DESIGNATION.		TERMINAL UNIT TAG
	EXISTING DUCTWORK / PIPING TO REMAIN		REHEAT DESIGNATION
	EXISTING DUCTWORK / PIPING TO BE REMOVED.		BOX DESIGNATION
	SECTION REFERENCE NUMBER.		AHU OR MAU DESIGNATION
	SECTION SHEET NUMBER		WORK NOTE DESIGNATION.
			POINT OF CONNECTION.
			HUMIDITY SENSOR
			TEMPERATURE SENSOR.
			TEMPERATURE TRANSMITTER
			CARBON MONOXIDE SENSOR.
			CARBON DIOXIDE SENSOR.
			INDOOR AIR QUALITY SENSOR.
			OCCUPANCY SENSOR.

SHEET METAL FITTINGS AND EQUIPMENT

	SUPPLY AIR DIFFUSER		DUCT TRANSITION
	RETURN AIR GRILLE		DUCT TRANSITION
	EXHAUST AIR GRILLE		SQUARE TO ROUND DUCT TRANSITION
	LINEAR SLOT DIFFUSER		AIRFLOW - SUPPLY
	SUPPLY DUCT UP		AIRFLOW - RETURN OR EXHAUST
	SUPPLY DUCT DOWN		MANUAL VOLUME DAMPER
	RETURN DUCT UP		MANUAL VOLUME DAMPER WITH REMOTE OPERATION
	RETURN DUCT DOWN		COMBINATION FIRE/SMOKE DAMPER.
	EXHAUST DUCT UP		FIRE DAMPER.
	EXHAUST DUCT DOWN		SMOKE DAMPER.
			MOTORIZED CONTROL DAMPER

PIPING DESIGNATIONS AND FITTINGS

	HWS	HEATING WATER SUPPLY		ISOLATION VALVE
	HWR	HEATING WATER RETURN		CHECK VALVE
	HRS	HEAT RECOVERY SUPPLY		PLUG VALVE
	HRR	HEAT RECOVERY RETURN		DYNAMIC VALVE
	CHWS	CHILLED WATER SUPPLY		TWO-WAY CONTROL VALVE
	CHWR	CHILLED WATER RETURN		THREE-WAY CONTROL VALVE
	CWS	CONDENSER WATER SUPPLY		BALANCING VALVE
	CWR	CONDENSER WATER RETURN		PRESSURE REDUCING VALVE
	LPS	LOW PRESSURE STEAM		STRAINER
	LPR	STEAM (CONDENSATE) RETURN		TEST PORT, UNION
	RL	REFRIGERANT LIQUID		THERMOMETER, PRESSURE GAUGE
	RS	REFRIGERANT SUCTION		WELL, MANUAL AIR VENT
	RHG	REFRIGERANT HOT GAS		PIPE DOWN AND PIPE TEE DOWN
	CD	CONDENSATE DRAIN		PIPE UP AND PIPE TEE UP
	PCD	PUMPED CONDENSATE DRAIN		PIPE CAP, BLIND FLANGE
	RH	RADIANT HEATING		PIPE ANCHOR/ALIGNMENT GUIDE
	SM	SNOWMELT		
		ARROW IN LINE INDICATES DIRECTION OF FLOW		

PIPING NOTATION

—X— PIPE TYPE (XXX)

XXX = GPM OF FLOW FOR PIPING

MECHANICAL DRAWING INDEX

SHEET NUMBER	SHEET TITLE					
		BID SET	PERMIT SET			
		07/20/2020	07/20/2020			
M0.0	MECHANICAL INDEX, LEGEND AND NOTES	○	●			
M0.1	MECHANICAL SCHEDULES	○	●			
M0.2	MECHANICAL SCHEDULES	○	●			
M0.3	MECHANICAL DIAGRAMS	○	●			
M0.4	MECHANICAL DIAGRAMS	○	●			
M0.5	MECHANICAL SPECIFICATIONS	○	●			
M1.0	MAIN FLOOR PLAN - MECHANICAL	○	●			
M2.0	ROOF PLAN - MECHANICAL	○	●			
TOTAL:		8	8			

DESIGN DATA

LOCATION:	GREELEY, CO	ALTITUDE:	4700 FT
		DENSITY RATIO:	0.842
ASHRAE CLIMATE ZONE:	5B		
OUTDOOR CONDITIONS:	SUMMER	DESIGN DB =	95 °F
	WINTER	COINCIDENT WB =	64 °F
		DESIGN DB =	-10 °F
INDOOR CONDITIONS:	SUMMER	DESIGN DB =	75 °F
	WINTER	DESIGN DB =	70 °F
CODES:	2018 INTERNATIONAL BUILDING CODE		
	2018 INTERNATIONAL MECHANICAL CODE		
	2018 INTERNATIONAL PLUMBING CODE		
	2018 IECC		
	ALL APPLICABLE LOCAL AMENDMENTS		
VENTILATION:	PER ASHRAE 62.1 AND 2018 IMC.		
ENVELOPE CHARACTERISTICS:	REFERENCE ARCHITECTURAL PLANS.		

CRIME LAB STORAGE BUILDING
2329 115TH AVE
Greeley, CO

MECHANICAL INDEX, LEGEND AND NOTES

DESIGNED
DRAWN
CHECKED
APPROVED

G2

DATE

DRAWING NO.
M0.0

REVISIONS

DATES

REGISTERED PROFESSIONAL ENGINEER
#38512
Gregory Peterson
01/24/2020

G2CE#2019086

CONSULTING ENGINEERS, INC.
1039 MAIN STREET
WINDSOR, CO 80550
TEL: 970-346-0151
GEE.COM

RS&A
ROBERT SHREVE ARCHITECTS & PLANNERS
MADISON & MAIN BUILDING
801 8TH STREET, SUITE 120
GREELEY, COLORADO 80631
970-346-0151 FAX 970-352-8761

ROOFTOP PACKAGE UNIT SCHEDULE

TAG	MANUF. / MODEL	LOCATION	SERVICE	VAV/ CV UNIT	EER / IEER (ARI)	AIR QUANTITIES IN CFM					FAN DATA						ELECT DATA						NATURAL GAS FIRED HEATING						DX COOLING						GENERAL		
						SUPPLY	UNIT RETURN	MIN VENT OA	MAX VENT OA	MAX ECON OA	SYSTEM	TYPE	SIZE / RPM	ESP / TSP ("WC)	VFD	BRAKE / MOTOR HP	VOLTS/ Ø	MCA	MOCP	E-PWR	NOTES	SEA LEVEL INPUT (MBH)	SITE OUTPUT (MBH)	CONTROL	EFF.	AIRSIDE DATA			NOTES	TOTAL (MBH)	SENSIBLE (MBH)	AIRSIDE DATA		NOTES	OPER. WEIGHT (LBS.)	ACCESSORIES	TAG
																										EAT DB ("F)	LAT DB ("F)	NOTES				COIL CAPACITY	COIL LAT ("F) DB/WB				
RTU 1	TRANE OAGD120A4	ROOF	OFFICE	VAV	10.5	3250	1250	2000	2000	3250	SUPPLY	BC PLENUM	1692	1.5 / 1.74	ECM	1.94 / 3.75	460/3	26.9	35	N/A	1, 2	350	235	10:1	80%	21	87.8	1, 2	122.9	105	87.3 / 63.4	52.3 / 51.5	66.4 / 56.8	1, 2, 3	3100	1, 2, 3, 4, 6, 7	RTU 1
RTU 2	TRANE YHC060F4	ROOF	STORAGE	CV	12.8 / 15.0	1980	1580	400	400	1980	SUPPLY	FC CENTRIFUGAL	1213	1.25 / 1.5	ECM	1.2 / 1.0	460/3	12	15	N/A	1, 2	130	108	1-STAGE	80%	54	102.7	1, 2	58.0	46.0	79 / 63	55.7 / 52.6	57.6 / 53.4	1, 2, 3	1150	1, 2, 3, 4, 5, 7	RTU 2
											EXHAUST	FC CENTRIFUGAL	1075	0.2	N	0.33		1.1 FLA	2.0 LRA																		
RTU 3	TRANE OABD036E4	ROOF	VAULT #1 & #2	CV	10.5	800	0	800	800	800	SUPPLY	BC PLENUM	1587	1.25/1.38	ECM	0.42 / 2.5	460/3	15.4	20	N/A	1, 2	100	67.2	5:1	80%	-10	68	1, 2	35.2	34.1	95 / 64	49.4 / 48.1	50.9 / 48.8	1, 2, 3	1500	1, 2, 4, 7	RTU 3
ACCESSORIES: 1) 14" FACTORY ROOF CURB. 2) FAR 30/30 MERV 8 FILTERS. 3) 100% MODULATING ECONOMIZER AND INTEGRAL ECONOMIZER FAULT DETECTION COMPLYING WITH 2018 IECC 4) VERTICAL (DOWN) DISCHARGE. 5) PROVIDE POWER EXHAUST. 6) PROVIDE RA SMOKE DETECTOR. 7) EC MOTOR.																NOTES: 1) SINGLE POINT POWER CONNECTION, 2) FUSED DISCONNECT PROVIDED BY ELECTRICAL CONTRACTOR.						NOTES: 1) INPUT GAS PRESSURE = 5" WC MIN TO 14" WC MAX. 2) STAINLESS STEEL HEAT EXCHANGER (MAT BELOW 45 DEGREES.						NOTES: 1) R-410A REFRIGERANT, 2) CAPACITIES INDICATED AT 95F AMBIENT AND 4700 FT ALTITUDE. 3) CAPACITIES ARE NET AND ACCOUNT FOR FAN HEAT.									

INFRA-RED TUBE HEATER SCHEDULE

TAG	MANUF.	MODEL	LOCATION	LENGTH	INPUT CAPACITY (MBH)	ELECTRICAL DATA		WEIGHT (LBS)	NOTES
						VOLTAGE / PHASE	MCA		
IRH 1	SCHWANK	S100-80	GARAGE 131	20'-0"	80	120/1	1.5 AMPS	150	1, 2, 3
IRH 2	SCHWANK	S100-80	GARAGE 131	30'-0"	80	120/1	1.5 AMPS	150	1, 2, 3
NOTES: 1) MIN GAS PRESSURE = 5" WC, MAX GAS PRESSURE = 14" WC. 2) PROVIDE WITH REFLECTOR AND END CAP. 3) HONEYWELL FOCUSPRO TH500 THERMOSTAT (24VC DIGITAL)									

GAS FIRED UNIT HEATER SCHEDULE

TAG	MANUF.	MODEL	LOCATION	MOUNTING	NATURAL GAS FIRED HEATING			AIRSIDE DATA		VENT (IN)	COMBUSTION AIR (IN)	ELECTRICAL				WEIGHT (LBS)	NOTES
					SEAL LEVEL INPUT (MBH)	SITE OUTPUT (MBH)	EFF.	AIRFLOW (CFM)	TEMP. RISE (°F)			VOLTS/Ø	FLA	MOCP	E-POWER		
UH-1	REZNOR	UDAS 76	GARAGE 119	CEILING	75	45	83%	960	60	4"	4"	120/1	3.3	15	N	80	1, 2, 3, 4, 5
NOTES: 1) 2-STAGE CONTROLLER. 2) MIN GAS PRESSURE = 5" WC, MAX GAS PRESSURE = 14" WC. 3) HIGH ALTITUDE KIT. 4) HONEYWELL FOCUSPRO TH500 THERMOSTAT (2 STAGE 24VC DIGITAL), THERMOSTAT GUARD W/ LOCKING COVER. 5) VERTICAL COMBUSTION AIR/VENT KIT W/ CONCENTRIC ADAPTER.																	

BUILDING AIR BALANCE SCHEDULE

EQUIPMENT	EA (CFM)	OA (CFM)
RTU-1		2,000
RTU-2		400
RTU-3 (DOAS)		800
EF-1	70	
EF-2	140	
EF-3	140	
EF-6 (DOAS)	800	
EF-7	600	
EF-8	80	
EF-9 (FUME HOOD)	400	
EF-10 (FUME HOOD)	300	
TRANSFER AIR FROM BUILDING		
TOTALS	2,530	3,200

SPLIT SYSTEM INDOOR UNIT SCHEDULE

TAG	MANUF.	MODEL	LOCATION	SERVICE	CONFIGURATION	AIR QUANTITIES		ELECTRICAL DATA			WEIGHT (LBS)	NOTES
						SUPPLY AIR (CFM)	OUTSIDE AIR (CFM)	VOLTAGE / PHASE	MCA	MOCP		
AC-1	MITSUBISHI	PKA-A24KA4	IT ROOM 133	IT ROOM	WALL MOUNTED	700	--	208/1	1	15	50	1, 2
NOTES: 1) PROVIDE WITH WALL MOUNTED 7-DAY PROGRAMMABLE THERMOSTAT. 2) PROVIDE WITH CONDENSATE PUMP												

SPLIT CONDENSING UNIT SCHEDULE

TAG	MANUF.	MODEL	LOCATION	SERVICE	AMBIENT TEMP	CAPACITY (MBH)		EER / SEER	ELECTRICAL DATA			WEIGHT (LBS)	NOTES
						COOLING	HEATING		VOLTAGE / PHASE	MCA	MOCP		
CU-1	MITSUBISHI	PUY-A24NH44	ROOF	AC-1	95	24	N/A	10.6 / 17.0	208/1	18	30	180	1, 2, 3, 4, 5, 6, 7
NOTES: 1) PROVIDE WITH INSULATED REFRIGERANT LINE SETS INSTALLED ACCORDING TO MANUFACTURER'S RECOMMENDATIONS. 2) LOW AMBIENT KIT TO 0F. 3) 4" HOUSEKEEPING PAD WITH 12" TALL QUICKSLING STAND. 4) WIND BAFFLE. 5) HAIL GUARD. 6) WINTER START CONTROL. 7) LIQUID LINE SOLENOID VALVE AND SIGHT GLASS.													

DIFFUSER, GRILLE AND REGISTER SCHEDULE

TAG	MANUF	MODEL	SYSTEM	SIZE	DESCRIPTION	CONSTRUCTION	OBD	FINISH	NOTES
A	TITUS	272-RL	SUPPLY	VARIES	3/4" SPACING, DOUBLE DEFLECTION.	ALUMINUM	Y	1, 2	A, B, C, D
B	TITUS	350-RL	RETURN	VARIES	3/4" SPACING, 35 DEG. DEFLECTION.	ALUMINUM	N	1, 2	A, B, C, D
C	TITUS	350-RL	EXHAUST	VARIES	3/4" SPACING, 35 DEG. DEFLECTION.	ALUMINUM	Y	1, 2	A, B, C, D
D	TITUS	TMS	SUPPLY	24x24	DROPPED LOUVER FACE	ALUMINUM	N	1	A, B, C, D
E	TITUS	TMS	SUPPLY	20x20	DROPPED LOUVER FACE	ALUMINUM	N	1	A, B, C, D
F	TITUS	TMS	SUPPLY	12x12	DROPPED LOUVER FACE	ALUMINUM	N	1	A, B, C, D
G	TITUS	PXP-AA	RETURN	24x24	PERFORATED FACE.	ALUMINUM	N	1	A, B, C
FINISHES: 1) FACTORY FINISH STANDARD OFF WHITE. 2) DIFFUSER TO BE FIELD PAINTED, COLOR TO BE DETERMINED BY ARCHITECT.					NOTES: A) SEE DRAWINGS FOR NECK SIZE AND AIR QUANTITY. B) PROVIDE FRAME COMPATIBLE WITH CEILING TYPE. C) REFERENCE ARCHITECTURAL PLANS FOR FINAL DIFFUSER LOCATIONS. D) PROVIDE SQUARE-TO-ROUND TRANSITION AT DIFFUSER CONNECTION AS REQUIRED.				

CRIME LAB STORAGE BUILDING

2329 115TH AVE

MECHANICAL SCHEDULES

WELD COUNTY

CRIME LAB STORAGE BUILDING

GREELEY, CO.

DESIGNED

DRAWN

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APPROVED

G2

G2

G2

DATES

REVISIONS

DATE

DRAWING NO.

M0.1

CONSULTING ENGINEERS, INC.

1038 MAIN STREET

WINDSOR, CO 80550

TEL: 970-346-0151

FAX: 970-346-8761

G2CE#2019086

RS&A

ROBERT SHREVE ARCHITECTS & PLANNERS

MADISON & MAIN BUILDING

801 8TH STREET, SUITE 120

GREELEY, COLORADO 80631

970-346-0151 FAX 970-352-8761

FAN SCHEDULE

TAG	MANUF.	MODEL	LOCATION	SERVICE	FAN DATA						MOTOR DATA					SOUND POWER DATA	OPER. WEIGHT (LBS)	NOTES	ACCESSORIES	TAG
					AIR QTY. (CFM)	ESP ("W.C.)	FEG	TYPE	RPM	MOUNT	DRIVE	HP	VOLTSØ	VFD	E- POWER					
EF-1	COOK	GC-128	UNISEX RR 104	TOILET ROOM	70	0.125	N/A	CENTRIFUGAL	750	CEILING	DIRECT	25 WATT	277/1	N	N/A	0.9	20	A	1, 3, 6	EF-1
EF-2	COOK	GC-148	UNISEX RR 116	TOILET ROOM	140	0.125	N/A	CENTRIFUGAL	1075	CEILING	DIRECT	45 WATT	277/1	N	N/A	2.65	25	A	1, 3, 6	EF-2
EF-3	COOK	GC-148	UNISEX RR 123	TOILET ROOM	140	0.125	N/A	CENTRIFUGAL	1075	CEILING	DIRECT	45 WATT	277/1	N	N/A	2.5	25	A	1, 3, 6	EF-3
EF-4	COOK	10XW40D17EC	WEST WALL GARAGE 119	GARAGE 119	550	0.15	N/A	PROPELLER	1725	WALL	DIRECT	1/4	120/1	N	N/A	7.9	50	B	1, 3, 4, 5	EF-4
EF-5	COOK	14XW32D17EC	WEST WALL GARAGE 131	GARAGE 131	1720	0.25	N/A	PROPELLER	1725	WALL	DIRECT	1/4	120/1	N	N/A	13.5	60	B	1, 3, 4, 5	EF-5
EF-6	COOK	101C17DEC	ROOF	VAULT #1 AND #2	800	0.25	N/A	CENTRIFUGAL	1725	ROOF	DIRECT	1/4	120/1	N	N/A	7.1	75	D	1, 2, 3	EF-6
EF-7	COOK	90C17DEC	ROOF	DRY ROOMS	600	0.25	N/A	CENTRIFUGAL	1725	ROOF	DIRECT	1/4	120/1	N	N/A	6.1	90	E	1, 2, 3	EF-7
EF-8	COOK	GC-128	MECH/JAN 109	JANITOR ROOM	80	0.125	N/A	CENTRIFUGAL	750	CEILING	DIRECT	25 WATT	120/1	N	N/A	0.9	20	F	1, 3, 6	EF-8
NOTES: A) INTERLOCK FAN OPERATION W/LIGHTS/OCCUPANCY SENSOR. B) INTERLOCK FAN OPERATION W/MOTORIZED DAMPER @ OA INTAKE LOUVER AND CO/NO2 SENSOR. C) INTERLOCK FAN OPERATION W/OWNER PROVIDED FUME HOOD. D) INTERLOCK W/ RTU-3 OPERATION. E) INTERLOCK W/RTU1-3 VAV BOX CONTROL. F) OPERATE CONTINUOUSLY.									ACCESSORIES: 1) FACTORY PROVIDED DISCONNECT SWITCH. 2) 22" ROOF CURB. 3) EC MOTOR. 4)MOTORIZED DAMPER INTERLOCKED TO FAN OPERATION AND OA INTAKE. 5) WIREGUARD ON MOTOR SIDE, WALL COLLAR AND WEATHERHOOD. 6) BACKDRAFT DAMPER.											

ELECTRIC HEATER SCHEDULE

TAG	MANUF.	MODEL	LOCATION	MOUNTING	SIZE	KW	AIRFLOW			ELECTRICAL		E-POWER	NOTES	TAG
							CFM	EAT	LAT	VOLTS/Ø	AMPS			
EWH-1	QMARK	AWH4408F	VESTIBULE 101	WALL	15-3/4" W x 19-5/16" H	4	--	--	--	208/1	19.2	N	1, 2, 3	EWH-1
EWH-2	QMARK	AWH4408F	VESTIBULE 117	WALL	15-3/4" W x 19-5/16" H	4	--	--	--	208/1	19.2	N	1, 2, 3	EWH-2
NOTES: 1) THERMAL CUTOUT SWITCH. 2) TAMPER PROOF INTEGRAL THERMOSTAT. 3) PROVIDE WITH WALL BRACKET														

SINGLE DUCT VARIABLE AIR VOLUME BOX SCHEDULE (ALL VAV BOX FRAMES MAY NOT BE USED)

MFR.	MODEL	INLET SIZE	CFM CONTROL RANGE	AIR P.D. ("WC)	MIN HEATING CFM	MAX HEATING CFM	COIL HEATING EAT (F)	COIL HEATING LAT (F)	ELECTRIC HEATING KW	CONTROL	ELECTRICAL		ACCESSORIES
											VOLTS/ø	OPTIONAL STANDBY POWER	
TITUS	DESV	6"	45-350	0.1	N/A	N/A	N/A	N/A	N/A	N/A	277/1	N	1, 2, 3
TITUS	DESV	6"	45-350	0.1	45	175	55	90	RE FLOOR PLANS	SCR	480/3	N	1, 2, 3, 4, 5
TITUS	DESV	8"	90-650	0.19	90	325	55	90	RE FLOOR PLANS	SCR	480/3	N	1, 2, 3, 4, 5
TITUS	DESV	10"	145-1050	0.21	145	525	55	90	RE FLOOR PLANS	SCR	480/3	N	1, 2, 3, 4, 5
TITUS	DESV	12"	190-1500	0.53	190	750	55	90	RE FLOOR PLANS	SCR	480/3	N	1, 2, 3, 4, 5
TITUS	DESV	14"	300-2400	0.27	300	1200	55	90	RE FLOOR PLANS	SCR	480/3	N	1, 2, 3, 4, 5
TITUS	DESV	16"	385-3200	0.31	385	1600	55	90	RE FLOOR PLANS	SCR	480/3	N	1, 2, 3, 4, 5
NOTES: 1) 24 VOLT STEP DOWN TRANSFORMER 2) SINGLE POINT ELECTRICAL CONNECTION 3) THERMAL OVERLOADS 4) AIR FLOW SWITCH 5) MANUAL AND AUTOMATIC HIGH TEMPERATURE SAFETY													

CRIME LAB STORAGE BUILDING
2329 115TH AVE
Greeley, CO

MECHANICAL SCHEDULES

WELD COUNTY

CRIME LAB STORAGE BUILDING
2329 115TH AVE
Greeley, CO

DESIGNED G-2
DRAWN
CHECKED
APPROVED G-2

DATE
DRAWING NO:
M0.2

REVISIONS

DATES

REGISTERED
PROFESSIONAL ENGINEER
#38512
Gregory Petersen

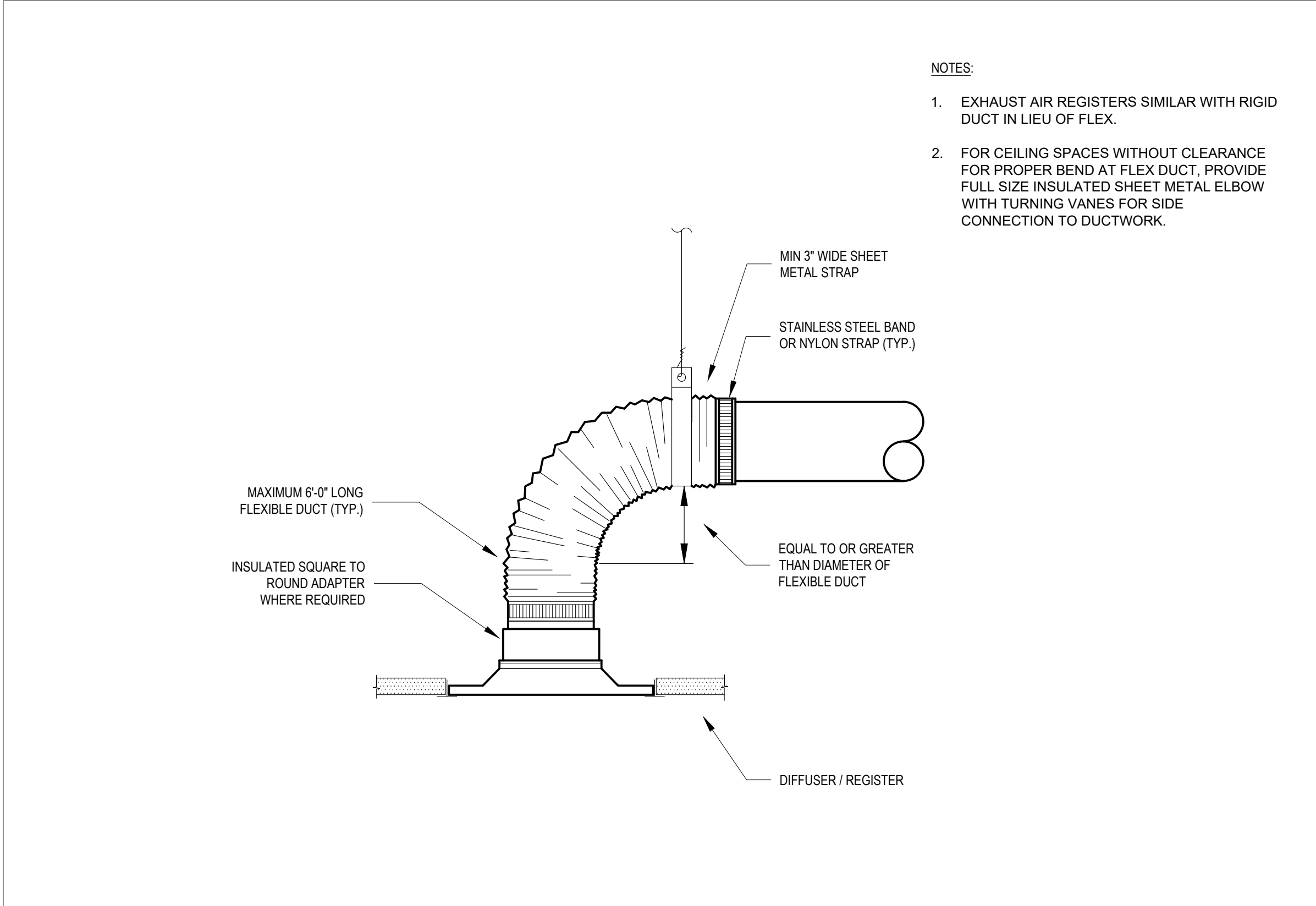
01/24/2020

G2CE#2019086

CONSULTING ENGINEERS, INC.
1039 MAIN STREET
WINDSOR, CO 80550
TEL: 970-346-0151
G2CE.COM

RS&A

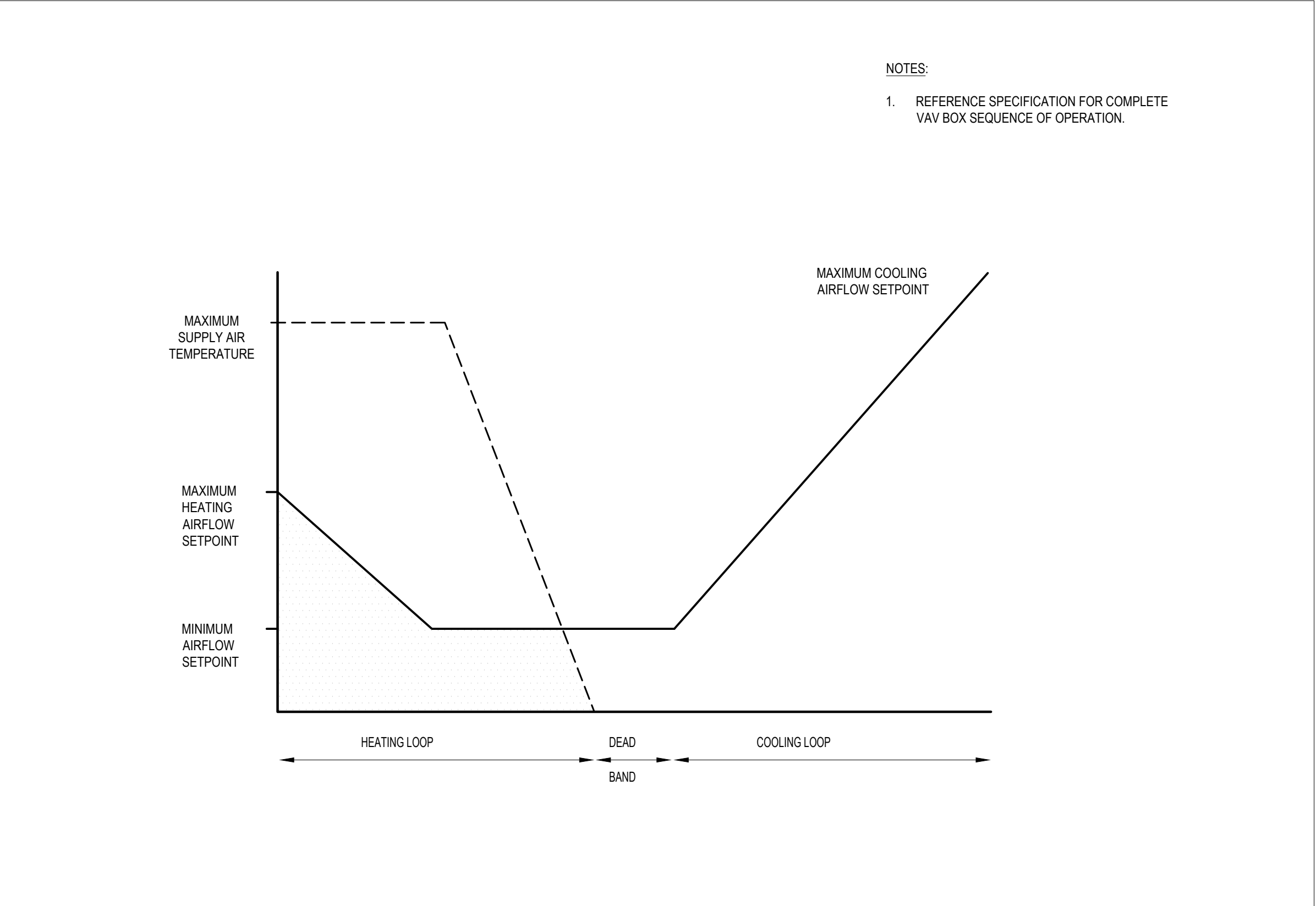
ROBERT SHREVE ARCHITECTS & PLANNERS
MADISON & MAIN BUILDING
801 8TH STREET, SUITE 120
GREELEY, COLORADO 80631
970-346-0151 FAX 970-352-8761



CEILING DIFFUSER INSTALLATION

N.T.S.

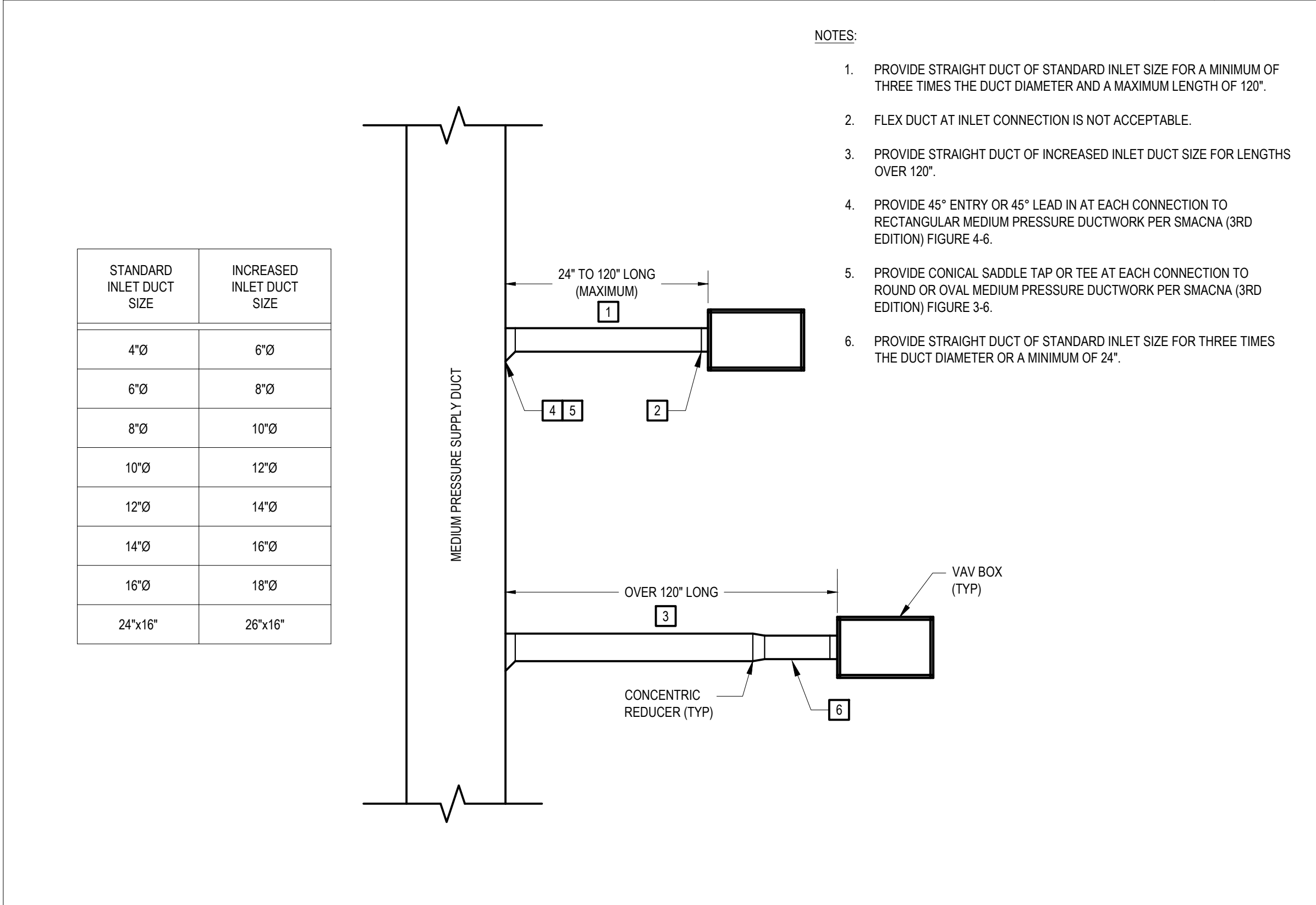
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VAV BOX CONTROL LOGIC

N.T.S.

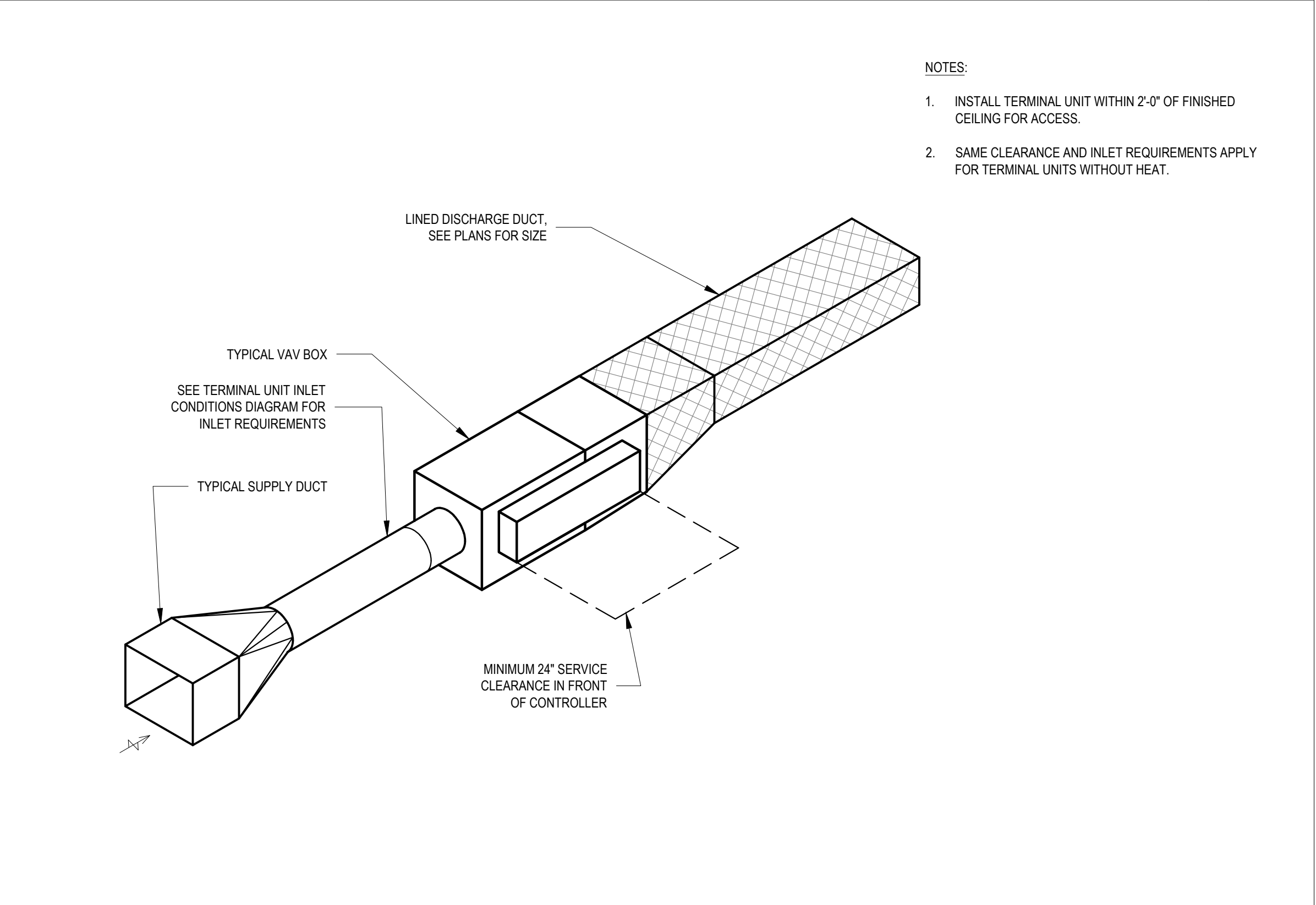
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M0.3



VAV TERMINAL UNIT INLET CONDITIONS

N.T.S.

4
M0.3



VAV TERMINAL UNIT WITH ELECTRIC REHEAT COIL

N.T.S.

2
M0.3

MECHANICAL DUCT SYSTEM NOTES

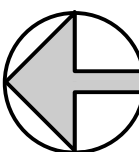
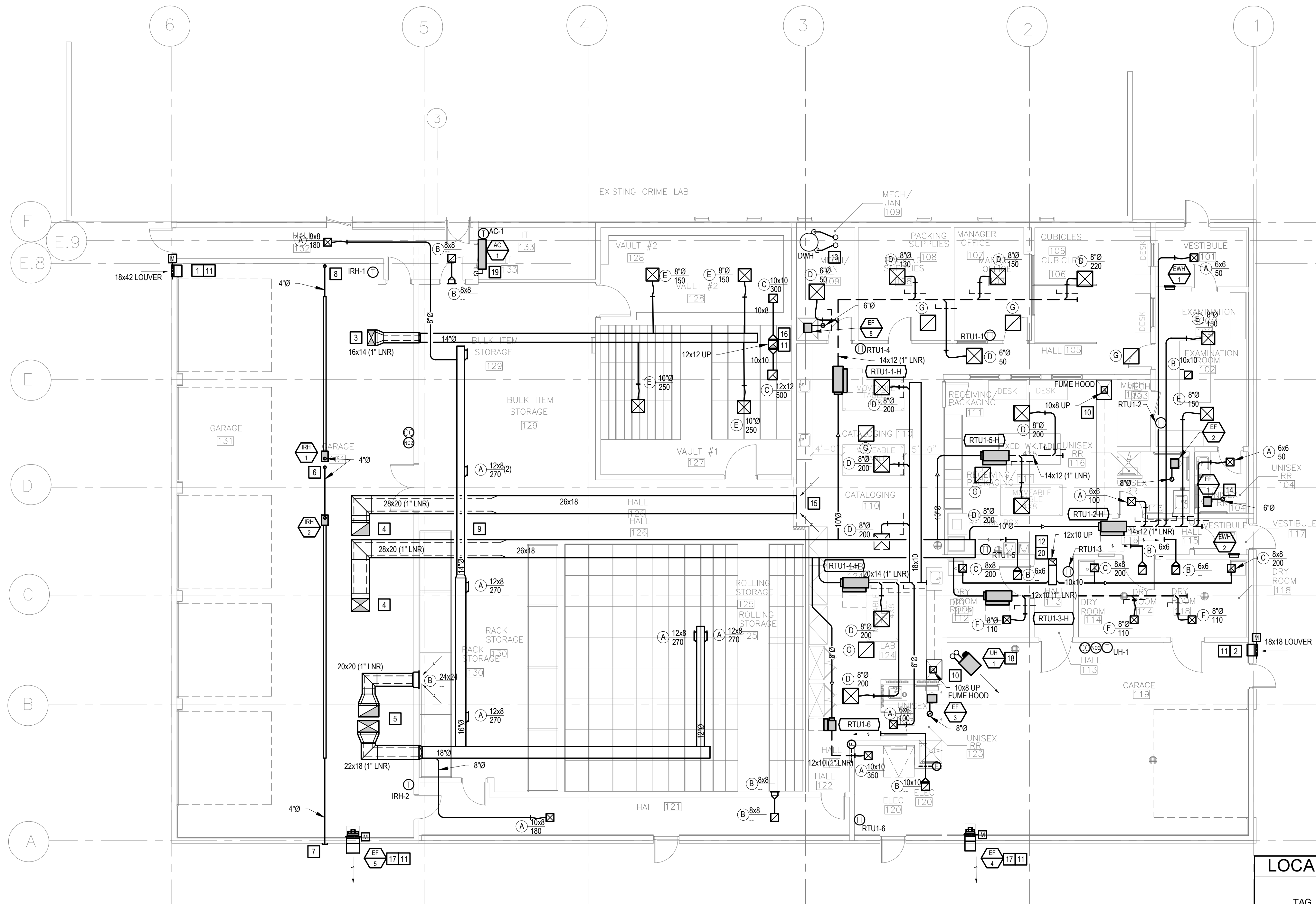
1. DUCT SIZES AS INDICATED ON THE DRAWINGS ARE OUTSIDE SHEET METAL MEASUREMENTS. WHERE DUCT LINER IS USED THE DUCTWORK DIMENSIONS ACCOMMODATE THE DUCT LINER.
2. CONTRACTOR SHALL CONSTRUCT AND INSTALL DUCTWORK PER SMACNA STANDARDS.
3. EXHAUST DUCTWORK CONNECTED TO FUME EXHAUST HOODS AND EXHAUST GRILLES SHALL HAVE WELDED SEAMS AND EITHER WELDED OR FLANGED JOINTS. FLANGED JOINTS SHALL BE WATERTIGHT WITH EPDM GASKETS. SLOPE DUCTWORK BACK TOWARDS HOOD/EXHAUST GRILLE AT 1% SLOPE.
4. BRANCH DUCT CONNECTIONS TO DIFFUSERS SHALL BE THE SAME SIZE AS THE DIFFUSER NECK UNLESS NOTED OTHERWISE.
5. REFERENCE VAV INSTALLATION DIAGRAM FOR INLET REQUIREMENTS. ALL VAV BOXES SHALL BE LOCATED 2'-0" ABOVE THE FINISHED CEILING TO ALLOW MAINTENANCE ACCESS.
6. SPIN-IN FITTINGS TO VAV BOXES AND DIFFUSERS SHALL BE CONICAL TYPE (EXCEPT LOCATIONS WHERE LISTED DUCT HEIGHT DOES NOT ACCOMMODATE).
7. DAMPERS: SINGLE BLADE TYPE VOLUME DAMPERS SHALL BE INSTALLED AT ALL DIFFUSER TAKEOFFS AND WHERE REQUIRED FOR PROPER BALANCING. HANDLE SHALL BE VISIBLE THROUGH INSULATION.
8. REMOTE DAMPERS: PROVIDE A REMOTE DAMPER ACTUATOR FOR LOCATIONS WHERE DAMPERS ARE NOT ACCESSIBLE. WHERE REMOTE DAMPER ACTUATORS ARE PROVIDED, COORDINATE LOCATION OF REMOTE DAMPER ESCUTCHEON PLATE AND COVER WITH ARCHITECT.
9. PROVIDE A MINIMUM 12" LONG RED RIBBON LOCATOR ON VOLUME DAMPER VALVE HANDLES.
10. PROVIDE ACCESS DOOR IN DUCTWORK UPSTREAM OF EACH DUCT-MOUNTED COIL, HUMIDIFIER, SMOKE DETECTOR, AND COMBINATION FIRE/SMOKE DAMPER.
11. SPIRAL EXPOSED DUCTWORK SHALL BE GALVANIZED SHEET STEEL SPIRAL. PROVIDE MILL PHOSPHATIZED FINISH THAT IS FREE FROM VISUAL IMPERFECTIONS, INCLUDING PITTING, SEAM MARKS, ROLLER MARKS, STAINS, DENTS AND DISCOLORATIONS, INCLUDING THOSE THAT WOULD IMPAIR PAINTING.
12. RECTANGULAR DUCT TURNS/ELBOWS: ALL 90 DEGREE ELBOWS SHALL BE LONG RADIUS ELBOWS OR SHALL HAVE TURNING VANES CONSISTING OF SINGLE BLADE DUCT VANES WITH 2-1/2 INCH BLADE SPACING.
13. INSULATED FLEXIBLE DUCT MAY BE USED FOR THE CONNECTION TO SUPPLY AIR OUTLETS/DIFFUSERS PROVIDED THE FLEXIBLE CONNECTION DOES NOT EXCEED 6 LINEAR FEET IN LENGTH. INSTALL DUCTS FULLY EXTENDED. DO NOT INSTALL IN THE COMPRESSED STATE OR USE EXCESS LENGTH.

DUCT SCHEDULE						
	MATERIAL	PRESSURE CLASS	MINIMUM SMACNA SEAL CLASS	SMACNA LEAKAGE CLASS FOR RECT.	SMACNA LEAKAGE CLASS FOR ROUND	DUCT SYSTEM NOTES
SUPPLY AND RETURN DUCTS						
UPSTREAM AND DOWNSTREAM OF FANS WITH GREATER THAN 2" ESP	GALVANIZED STEEL	4-INCH WG	A	6	3	
	UPSTREAM OF VAV BOXES	GALVANIZED STEEL	4-INCH WG	A	6	
DOWNSTREAM OF VAV BOXES	GALVANIZED STEEL	4-INCH WG	B	24	12	
UPSTREAM AND DOWNSTREAM OF FANS WITH LESS THAN 2" ESP	GALVANIZED STEEL	4-INCH WG	B	24	12	ALL EXTERIOR DUCTWORK TO BE SEAL CLASS A
EXHAUST DUCTS						
UPSTREAM AND DOWNSTREAM OF FANS WITH GREATER THAN 2" ESP	GALVANIZED STEEL	4-INCH WG	B	24	12	
	UPSTREAM AND DOWNSTREAM OF FANS WITH LESS THAN 2" ESP	GALVANIZED STEEL	2-INCH WG	B	24	
FUME HOOD EXHAUST	304 STAINLESS STEEL	3-INCH WG	AIRTIGHT/ WATER TIGHT	-	-	#3
INSULATION SCHEDULE						
	SYSTEM	LINER/WRAP	THICKNESS (IN)	DENSITY (LBS/FT3)	MINIMUM R-VALUE	NOTES
EXTERIOR DUCTWORK	SUPPLY	LINER	3"	1.5	R-12	
	RETURN	LINER	3"	1.5	R-12	
FIRST 20 FEET FROM AIR HANDLING UNITS, AND ROOFTOP UNITS	SUPPLY	LINER	2"	1.5	R-8	
	RETURN	LINER	2"	1.5	R-8	
RECTANGULAR DUCTWORK DOWNSTREAM OF VAV BOXES	SUPPLY	LINER	1"	1.5	R-4.2	
	RETURN	LINER	1"	1.5	R-4.2	
DUCTWORK IN UNCONDITIONED SPACE (VENTILATED ATTIC, BTWN ROOF AND INSULATED CEILING)	SUPPLY	WRAP	2"	0.75	R-6	WITH FSK FACING
	RETURN	WRAP	2"	0.75	R-6	WITH FSK FACING
DUCTWORK IN UNVENTILATED ATTIC (NO RA PLENUM)	SUPPLY	WRAP	2"	0.75	R-6	WITH FSK FACING
	RETURN	WRAP	2"	0.75	R-6	WITH FSK FACING
EXPOSED SPIRAL DUCT	SUPPLY/ RETURN	NONE	N/A	N/A	N/A	UNLESS NOTED ON PLANS
ALL OTHER SUPPLY DUCTWORK	SUPPLY	WRAP	1.5"	0.75	R-4.2	WITH FSK FACING
UNTREATED OUTSIDE AIR DUCT (ENCLOSED)	SUPPLY	WRAP	1.5"	0.75	R-4.2	WITH FSK FACING
	RETURN	WRAP	1.5"	0.75	R-4.2	WITH FSK FACING
TRANSFER DUCTS (AT FULL HEIGHT WALLS, RETURN BOOTS, ETC.)	RETURN	LINER	1"	1.5	R-4.2	
ALL OTHER RETURN WITHIN RA PLENUM	RETURN	NONE	N/A	N/A	N/A	
ALL OTHER EXHAUST	EXHAUST	NONE	N/A	N/A	N/A	

MECHANICAL SPECIFICATIONS

- | | | | | | |
|-------|--|--------|---|-------|--|
| 1.0 | GENERAL | 1.0 | STORE AND PROTECT FROM DAMAGE EQUIPMENT AND MATERIALS DELIVERED TO JOB SITE. COVER WITH WATERPROOF, TEAR-RESISTANT, HEAVY TARP OR POLYETHYLENE PLASTIC AS REQUIRED TO PROTECT FROM PLASTER, DIRT, PAINT, WATER, OR PHYSICAL DAMAGE. EQUIPMENT AND MATERIAL THAT HAS BEEN DAMAGED BY CONSTRUCTION ACTIVITIES WILL BE REJECTED, AND MECHANICAL CONTRACTOR IS OBLIGATED TO FURNISH NEW EQUIPMENT AND MATERIAL OF A LIKE KIND AS APPROVED BY OWNER. | 2.0 | STANDARDS AND ALL GOVERNING CODES. |
| 1.1 | MECHANICAL GENERAL REQUIREMENTS: | 1.1 | KEEP PREMISES UNDER CLEAN FROM FOREIGN MATERIAL CREATED DURING WORK PERFORMED UNDER THIS CONTRACT. PIPING, EQUIPMENT, ETC., SHALL HAVE A NEAT AND CLEAN APPEARANCE AT THE TERMINATION OF THE WORK. | 2.1 | PROVIDE ACCESS DOORS IN DUCTWORK FOR OBSERVATION AND MAINTENANCE OF ALL DAMPERS, COILS, FIVE DAMPERS, ETC. |
| 1.1.1 | PROVIDE ALL ITEMS FOR COMPLETE AND SUCCESSFUL OPERATION OF ALL MECHANICAL SYSTEMS. | 1.2 | PLUG OR CAP OPEN ENDS OF DUCTWORK AND PIPING SYSTEMS WHILE STORED OR INSTALLED DURING CONSTRUCTION WHEN NOT IN USE TO PREVENT THE ENTRANCE OF DEBRIS INTO THE SYSTEMS. | 2.1.3 | METAL AND GAUGE: UNLESS SPECIFIED OTHERWISE GALVANIZED IRON SHALL BE USED THROUGHOUT, FABRICATED AND INSTALLED SO THAT NO VIBRATING OR NOISE RESULTS. SHEET STEEL SHALL BE OF THE US STANDARD GAUGE AS RECOMMENDED IN THE LATEST EDITION OF SMACNA. |
| 1.1.2 | DEMONSTRATE THE OPERATION OF ALL SYSTEMS FOR THE OWNER AT A TIME AS DIRECTED BY THE OWNER. | 1.3 | KEEP THE MANUFACTURER PROVIDED PROTECTIVE COVERINGS ON FLOOR DRAINS, FLOOR SINKS, AND TRENCH DRAINS DURING THE CONSTRUCTION. REMOVE COVERINGS AT THE TERMINATION OF THE WORK AND POLISH EXPOSED SURFACES. | 2.1.5 | DUCT WORKMANSHIP: ALL DUCTWORK SHALL BE CONSTRUCTED AND ERECTED IN A WORKMANLIKE MANNER. DUCTS SHALL BE STRAIGHT AND SHOWN ON THE INSIDE WITH NEATLY FINISHED JOINTS, AIRTIGHT, AND SHALL BE FREE FROM VIBRATION UNDER ALL CONDITIONS OF OPERATION. THE INTERNAL ENDS OF SLIP JOINTS SHALL BE MADE IN THE DIRECTION OF AIRFLOW. DUCTS SHALL BE SECURELY ATTACHED TO THE BUILDING CONSTRUCTION IN AN APPROVED MANNER. |
| 1.1.3 | FIELD VERIFY ALL EXISTING CONDITIONS AND INCLUDE AN ALLOWANCE IN BID FOR REMOVAL AND/OR RELOCATION OF EXISTING PIPING, DUCTWORK, EQUIPMENT, ETC. AS INDICATED ON THE PLANS OR AS REQUIRED TO COORDINATE AND ADAPT NEW AND EXISTING SYSTEMS TO ALL OTHER WORK. CONTRACTORS SHALL VERIFY OPERATIONAL CONDITION OF ALL EQUIPMENT LOCATED WITHIN THE AREA PRIOR TO BEGINNING WORK. THE CONTRACTORS SHALL PREPARE A LIST OF DEFECTS AND DISCREPANCIES TO BE SUBMITTED TO THE MANAGER/OWNER PRIOR TO THE START OF CONSTRUCTION. THIS LIST SHALL ALSO INCLUDE ANY ITEMS THAT WERE NOT OR COULD NOT BE VERIFIED. | 1.4 | OPERATION AND MAINTENANCE INSTRUCTIONS: | 2.1.6 | DUCT TURNS: ALL 90 DEGREE ELBOWS UP TO 18 INCHES WIDE AND ALL 45 DEGREE ELBOWS SHALL HAVE AN INSIDE RADIUS OF NOT LESS THAN ONE HALF THE WIDTH OF THE DUCT. 90 DEGREE ELBOWS WITH TURNING VANES SHALL BE USED CONSISTING OF SINGLE BLADE DUCT VANES WITH 2.5 INCH BLADE SPACING. |
| 1.1.4 | REPLACE ANY WORK OR MATERIAL INSTALLED OR FURNISHED UNDER THIS CONTRACT WHICH DEVELOPS DEFECTS, EXCEPT FOR NORMAL WEAR, WITHIN ONE (1) YEAR AFTER CERTIFICATE OF COMPLETION IS SECURED. RETURN DEFECTIVE MATERIALS TO OWNER. | 1.5 | COLLECT AND COMPILE A COMPLETE BROCHURE OF FIXTURES, MATERIALS, AND EQUIPMENT FURNISHED AND INSTALLED ON THIS PROJECT. INCLUDE OPERATIONAL AND MAINTENANCE INSTRUCTIONS, MANUFACTURER'S CATALOG SHEETS, WIRING DIAGRAMS, PARTS LISTS, APPROVED SHOP DRAWINGS, AND DESCRIPTIVE LITERATURE FURNISHED BY THE MANUFACTURER. INCLUDE AN INSIDE COVER SHEET THAT LISTS THE PROJECT NAME, DATE, OWNER, ARCHITECT, ENGINEER, GENERAL CONTRACTOR, SUBCONTRACTOR, AND AN INDEX OF CONTENTS. | 2.1.7 | DAMPERS: SINGLE BLADE TYPE VOLUME DAMPERS SHALL BE INSTALLED AT DIFFUSER TAKEOFFS, AND WHERE REQUIRED FOR PROPER BALANCING. HANDLE SHALL BE VISIBLE THROUGH INSULATION. |
| 1.1.5 | CONTRACTOR SHALL OBTAIN OWNER OR BUILDING MANAGEMENT APPROVAL BEFORE CUTTING ANY FLOOR OR CEILING MATERIALS TO THE SATISFACTION OF THE SPECIFICATIONS AND DRAWINGS FOR THE PROJECT ARE COMPLEMENTARY, AND PORTIONS OF THE WORK DESCRIBED IN ONE, SHALL BE PROVIDED AS IF DESCRIBED IN BOTH. IN THE EVENT OF DISCREPANCIES, NOTIFY THE ENGINEER AND/OR OWNER AND REQUEST CLARIFICATION PRIOR TO PROCEEDING WITH THE WORK INVOLVED. | 1.6 | WARRANTIES: | 2.1.8 | GRILLES, REGISTERS, AND DIFFUSERS: PROVIDE SUPPLY, RETURN, AND EXHAUST REGISTERS AND DIFFUSERS AS NOTED ON THE EQUIPMENT SCHEDULE FOR THE TYPE, SIZE, MATERIAL, AND FINISH. |
| 1.1.6 | THE SPECIFICATIONS AND DRAWINGS FOR THE PROJECT ARE COMPLEMENTARY, AND PORTIONS OF THE WORK DESCRIBED IN ONE, SHALL BE PROVIDED AS IF DESCRIBED IN BOTH. IN THE EVENT OF DISCREPANCIES, NOTIFY THE ENGINEER AND/OR OWNER AND REQUEST CLARIFICATION PRIOR TO PROCEEDING WITH THE WORK INVOLVED. | 1.7 | 1.7.1 SUBMIT COPIES OF LITERATURE BOUND IN APPROVED BINDERS TO THE ARCHITECT AND OWNER AT THE TERMINATION OF THE WORK. PAPER CLIPS, STAPLES, RUBBER BANDS, AND METAL RINGS OR WRAPS ARE NOT CONSIDERED APPROVED BINDERS. FINAL APPROVAL OF MECHANICAL AND PLUMBING SYSTEMS WILL BE WITHHELD UNTIL THIS EQUIPMENT BROCHURE IS DEEMED COMPLETE BY THE ARCHITECT, ENGINEER, AND OWNER. | 2.1.9 | FLEX DUCT SHALL BE OWENS-CORNING FOIL-BACK HIGH QUALITY UL APPROVED. PLASTIC WRAPPED FLEX DUCT IS NOT ACCEPTABLE. |
| 1.2 | INSPECTION OF SITE: | 1.8 | 1.8.1 WARRANT EACH SYSTEM AND EACH ELEMENT THEREOF AGAINST ALL DEFECTS DUE TO FAULTY WORKMANSHIP, DESIGN OR MATERIAL FOR A PERIOD OF 12 MONTHS FROM DATE OF SUBSTANTIAL COMPLETION, UNLESS SPECIFIC ITEMS ARE NOTED TO CARRY A LONGER WARRANTY IN THE CONSTRUCTION DOCUMENTS OR MANUFACTURER'S STANDARD WARRANTY EXCEEDS 12 MONTHS. REMEDY ALL DEFECTS, OCCURRING WITHIN THE WARRANTY PERIOD(S), AS STATED IN THE GENERAL CONDITIONS AND DIVISION 1. | 2.2 | INSULATION: |
| 1.2.1 | PRIOR TO SUBMITTING BID, VISIT THE SITE OF THE PROPOSED WORK AND BECOME FULLY INFORMED AS TO THE CONDITIONS UNDER WHICH THE WORK IS TO BE DONE. FAILURE TO DO SO WILL NOT BE CONSIDERED SUFFICIENT JUSTIFICATION TO REQUEST OR OBTAIN EXTRA COMPENSATION OVER AND ABOVE THE CONTRACT PRICE. | 1.8.1 | 1.8.2 WARRANTIES SHALL INCLUDE LABOR AND MATERIAL. MAKE REPAIRS OR REPLACEMENTS WITHOUT ANY ADDITIONAL COSTS TO THE OWNER. | 2.2.1 | ALLOWABLE MANUFACTURERS:
JOHNS-MANVILLE CORP.
MANSON INSULATION CO.
OWENS-CORNING
OR APPROVED EQUAL. |
| 1.3 | MATERIAL AND WORKMANSHIP: | 1.8.2 | 1.8.3 PERFORM THE REMEDIAL WORK PROMPTLY, UPON WRITTEN NOTICE FROM THE ARCHITECT OR OWNER. | 2.2.2 | DUCT INSULATION TO BE FIBERGLASS INSULATION, WITH VAPOR BARRIER, UL APPROVED, CLASS I AIR DUCT. |
| 1.3.1 | PROVIDE NEW MATERIAL, EQUIPMENT, AND APPARATUS UNDER THIS CONTRACT UNLESS OTHERWISE STATED HEREIN, OF BEST QUALITY NORMALLY USED FOR THE PURPOSE IN GOOD COMMERCIAL PRACTICE, AND FREE FROM ANY DEFECTS. MODEL NUMBERS LISTED IN THE SPECIFICATIONS OR SHOWN ON THE DRAWINGS ARE NOT NECESSARILY INTENDED TO DESIGNATE THE REQUIRED TRIM, WRITTEN DESCRIPTIONS OF THE TRIM GOVERN MODEL NUMBERS. | 1.8.3 | 1.8.4 AT THE TIME OF SUBSTANTIAL COMPLETION, DELIVER TO THE OWNER ALL WARRANTIES, IN WRITING AND PROPERLY EXECUTED, INCLUDING TERM LIMITS FOR WARRANTIES EXTENDING BEYOND THE ONE YEAR PERIOD. EACH WARRANTY INSTRUMENT BEING ADDRESSED TO THE OWNER AND STATING THE COMMENCEMENT DATE AND TERM. | 2.2.3 | SEE DUCT SYSTEM NOTES AND INSULATION SCHEDULE FOR LOCATIONS AND INSULATION THICKNESSES. |
| 1.3.2 | WORK PERFORMED UNDER THIS CONTRACT SHALL PROVIDE A NEAT AND "WORKMANLIKE" APPEARANCE TO THE SATISFACTION OF THE ARCHITECT AND ENGINEER. WORKMANSHIP SHALL BE THE FINEST POSSIBLE BY EXPERIENCED MECHANICS. INSTALLATIONS SHALL COMPLY WITH APPLICABLE CODES AND LAWS. | 1.9 | CUTTING AND PATCHING: | 2.3 | ALL INSULATION SHALL MEET ALL CODE REQUIREMENTS AND FIRE AND SMOKE HAZARD RATINGS OF ASTM E-84, NATIONAL FIRE PROTECTION ASSOCIATION 225 AND UL 723, NOT EXCEEDING FLAME SPREAD 25 AND SMOKE DEVELOPED 50 RATINGS. |
| 1.3.3 | THE COMPLETE INSTALLATION SHALL FUNCTION AS DESIGNED AND INTENDED WITH RESPECT TO EFFICIENCY, CAPACITY, NOISE LEVEL, ETC. ABNORMAL NOISE CAUSED BY RATTLING EQUIPMENT, PIPING, DUCTS, AIR DEVICES, AND SQUEAKS IN ROTATING COMPONENTS WILL NOT BE ACCEPTABLE. IN GENERAL, MATERIALS AND EQUIPMENT SHALL BE OF COMMERCIAL SPECIFICATION GRADE IN QUALITY. LIGHT DUTY AND RESIDENTIAL TYPE EQUIPMENT WILL NOT BE ACCEPTED. | 1.9.1 | 1.9.1 PERFORM CUTTING OF WALLS, FLOORS, CEILINGS, ETC. AS REQUIRED TO INSTALL WORK UNDER THIS SECTION. OBTAIN PERMISSION FROM THE ARCHITECT PRIOR TO CUTTING. DO NOT CUT OR DISTURB STRUCTURAL MEMBERS WITHOUT PRIOR APPROVAL FROM THE ARCHITECT. CUT HOLES AS SMALL AS POSSIBLE. GENERAL CONTRACTOR SHALL PATCH WALLS, FLOORS, ETC. AS REQUIRED BY WORK UNDER THIS SECTION. PATCHING SHALL MATCH THE ORIGINAL MATERIAL AND CONSTRUCTION. REPAIR AND REFINISH AREAS DISTURBED BY WORK TO THE CONDITION OF ADJOINING SURFACES IN A MANNER SATISFACTORY TO THE ARCHITECT. | 2.3.1 | HVAC PIPING. |
| 1.3.4 | REMOVE FROM THE PREMISES WASTE MATERIAL PRESENT AS A RESULT OF WORK, INCLUDING CARTONS, CRATING, PAPER, STICKERS, AND/OR EMISSION MATERIALS NOT USED IN BACKFILLING, ETC. CLEAN EQUIPMENT INSTALLED UNDER THIS CONTRACT TO PRESENT A NEAT AND CLEAN INSTALLATION AT THE TERMINATION OF THE WORK. | 1.10 | ROUGH-IN: | 2.3.2 | CONDENSATE EQUIPMENT DRAIN TO BE COPPER WATER TUBE, TYPE "L" HARD DRAWN, WROUGHT COPPER FITTINGS, SOLDER JOINTS, USING 95-S SOLDER. |
| 1.3.5 | REPAIR OR REPLACE PROPERTY DAMAGED AS A RESULT OF WORK PERFORMED UNDER THIS CONTRACT TO THE SATISFACTION OF AUTHORITIES AND REGULATIONS HAVING JURISDICTION. | 1.10.1 | 1.10.1 COORDINATE WITHOUT DELAY ROUGHING-IN WITH GENERAL CONSTRUCTION. CONCEAL PIPING AND CONDUIT ROUGH-IN EXCEPT IN UNFINISHED AREAS AND WHERE OTHERWISE SHOWN. | 2.3.3 | PROOF SYSTEM, LEVER HANDLE AND ADJUSTABLE BALANCING MEMORY STOPS, SOLDER OR TREADED ENDS. PROVIDE EXTENDED STEMS FOR INSULATED PIPING. |
| 1.4 | COORDINATION: | 1.11 | STRUCTURAL STEEL: | 2.4 | TESTING, ADJUSTING, AND BALANCING: |
| 1.4.1 | COORDINATE WORK WITH THAT OF OTHER TRADES SO THAT THE VARIOUS COMPONENTS OF THE SYSTEMS THAT ARE BEING INSTALLED AND/OR EXISTING SYSTEMS WILL BE INSTALLED AT THE PROPER TIME, WITH THE AVAILABLE SPACE, AND WILL ALLOW PROPER SERVICE ACCESS TO THOSE ITEMS REQUIRING MAINTENANCE. COMPONENTS WHICH ARE INSTALLED WITHOUT REGARD TO THE ABOVE SHALL BE RELOCATED AT NO ADDITIONAL COST TO THE OWNER. | 1.11.1 | 1.11.1 STRUCTURAL STEEL USED FOR PIPE SUPPORTS, EQUIPMENT SUPPORTS, ETC., SHALL BE NEW, CLEAN, AND CONFORM TO ASTM DESIGNATION A-36. | 2.4.1 | SUBMIT TEST REPORTS FOR TESTING, ADJUSTING, AND BALANCING. |
| 1.4.2 | UNLESS OTHERWISE INDICATED, THE GENERAL CONTRACTOR WILL PROVIDE CHASES AND OPENINGS IN BUILDING CONSTRUCTION REQUIRED FOR INSTALLATION OF THE SYSTEMS SPECIFICITY STATED HEREIN. MECHANICAL CONTRACTOR SHALL FURNISH THE GENERAL CONTRACTOR WITH INFORMATION WHERE CHASES AND OPENINGS ARE REQUIRED, KEEP INFORMED AS TO THE WORK OF OTHER TRADES ENGAGED IN THE CONSTRUCTION OF THE PROJECT, AND EXECUTE WORK IN A MANNER AS TO NOT INTERFERE WITH OR DELAY THE WORK OF OTHER TRADES. | 1.11.2 | 1.11.2 SUPPORT PLUMBING AND MECHANICAL EQUIPMENT AND PIPING FROM THE BUILDING STRUCTURE. DO NOT SUPPORT PLUMBING EQUIPMENT FROM CEILINGS, OTHER MECHANICAL OR ELECTRICAL COMPONENTS, AND OTHER NON-STRUCTURAL ELEMENTS. | 2.4.2 | TOTAL SYSTEM BALANCE SHALL BE PERFORMED IN ACCORDANCE WITH NEBB OR AABC PROCEDURAL STANDARDS. |
| 1.4.3 | FINISHED DIMENSIONS SHALL TAKEN IN ACCORDANCE TO SCALE MEASUREMENTS AT THE BUILDING, AS VARIATIONS MAY OCCUR. MECHANICAL CONTRACTOR WILL BE HELD RESPONSIBLE FOR ERRORS THAT COULD HAVE BEEN AVOIDED BY PROPER CHECKING AND INSPECTION. | 1.12 | ACCESS DOORS: | 2.4.3 | BEFORE COMMENCING WORK, VERIFY THAT SYSTEMS ARE COMPLETE AND OPERABLE, WITH CLEAN FILTERS. |
| 1.4.4 | COORDINATE WORK WITH THAT OF OTHER TRADES SO THAT THE VARIOUS COMPONENTS OF THE SYSTEMS THAT ARE BEING INSTALLED AND/OR EXISTING SYSTEMS WILL BE INSTALLED AT THE PROPER TIME, WITH THE AVAILABLE SPACE, AND WILL ALLOW PROPER SERVICE ACCESS TO THOSE ITEMS REQUIRING MAINTENANCE. COMPONENTS | | | | |

WELD COUNTY					
CRIME LAB STORAGE BUILDING					
GREELY, CO.					
2329 115TH AVE					
MECHANICAL SPECIFICATIONS					
DRAWING NO. M0.5					
	DESIGNED DRAWN	G2			
	CHECKED				
	APPROVED	G2			
REVISIONS			DATES		
					
G2CE#2019086					
G2 CONSULTING ENGINEERS, INC. 1039 MAIN STREET UNIT G WINDSOR, COLORADO 80650 (970) 467-7400 G2CE.COM					
RSA ROBERT SHREV E ARCHITECTS & PLANNERS MADISON & MAIN BUILDING 801 8TH STREET, SUITE 120 GREELY, COLORADO 80631 970-346-0151 FAX 970-352-8761					



MAIN FLOOR PLAN - MECHANICAL

SCALE: 1/8" = 1'-0"

WORK NOTES: [A]

1. OA INTAKE W/DUCT EXTENSION, SIZE AS NOTED, PROVIDE MOTORIZED DAMPER COMPATIBLE WITH THE STATIONARY LOUVER (PROVIDED BY ARCHITECT), COORDINATE DAMPER OPERATION WITH EF-5.
2. OA INTAKE W/DUCT EXTENSION, SIZE AS NOTED, PROVIDE MOTORIZED DAMPER COMPATIBLE WITH THE STATIONARY LOUVER (PROVIDED BY ARCHITECT), COORDINATE DAMPER OPERATION WITH EF-4.
3. 16" x 14" SA DUCT UP TO RTU-3. TRANSITION DUCT AS REQUIRED FOR UNIT CONNECTION.
4. 28" x 20" SA AND RA DUCT UP TO RTU-1. TRANSITION DUCT AS REQUIRED FOR UNIT CONNECTION.
5. 22" x 18" SA, 20" x 20" RA DUCT UP TO RTU-2. TRANSITION AS REQUIRED FOR UNIT CONNECTION.
6. 4"Ø RADIANT TUBE HEATER AIR INTAKE. ROUTE UP THROUGH ROOF TO GOOSENECK TERMINATION.
7. 4"Ø RADIANT TUBE HEATER AIR VENT. ROUTE UP HORIZONTAL THROUGH WALL TO MANUFACTURER'S APPROVED SIDEWALL TERMINATION.
8. 4"Ø RADIANT TUBE HEATER AIR VENT. ROUTE UP THROUGH ROOF TO GOOSENECK TERMINATION.
9. SLOPE 14"Ø DUCT UP BETWEEN STRUCTURE TO ROUTE ABOVE 28"x20" SA DUCT. COORDINATE DUCT LOCATION AND ROUTE WITH STRUCTURE.
10. CONNECT 10x8 EA DUCT TO OWNER PROVIDED FUME HOOD BLOWER. ROUTE DUCT UP THROUGH ROOF WITH GOOSENECK.
11. PROVIDE MAN BARS AT EXTERIOR WALL/ROOF PENETRATION.
12. ROUTE 12x10 EA DUCT UP TO EF-7. TRANSITION DUCT AS REQUIRED TO MAKE FAN CONNECTION.
13. DOMESTIC WATER HEATER INTAKE AND VENT PIPING. ROUTE UP THROUGH ROOF TO CONCENTRIC VENT TERMINATION.
14. ROUTE TOILET ROOM EA DUCT UP THROUGH ROOF TO GOOSENECK TERMINATION. TYPICAL.
15. 25"x18" OPEN END RA DUCT.
16. ROUTE 12x12 EA DUCT UP TO EF-6. TRANSITION DUCT AS REQUIRED TO MAKE FAN CONNECTION.
17. WALL MOUNTED EF WITH MOTORIZED DAMPER. COORDINATE DAMPER OPERATION WITH OA INTAKE. COORDINATE FINAL EXHAUST FAN LOCATION WITH STRUCTURE, ARCHITECT/OWNER.
18. UNIT HEATER INTAKE AND VENT PIPING UP THROUGH ROOF TO CONCENTRIC TERMINATION.
19. PROVIDE CONDENSATE DRAIN PIPING FROM UNIT TO NEAREST FLOOR DRAIN/FLOOR SINK.

LOCAL VAV BOX SCHEDULE

TAG	INLET SIZE	ELEC HTG (KW)	MAX PRIMARY COOLING (CFM)	MIN PRIMARY (CFM)	MAX PRIMARY HEATING (CFM)	NOTES
RTU1-1-H	8"	3.5	600	150	300	A, B
RTU1-2-H	8"	2.5	450	115	225	A, B
RTU1-3-H	6"	2	330	85	165	A, B
RTU1-4-H	12"	6	1100	275	550	A, B
RTU1-5-H	8"	2.5	450	112.5	225	A, B
RTU1-6	6"	N/A	350	90	N/A	A, B

NOTES:

- A) LOCAL SCHEDULE FOR QUICK REFERENCE ONLY. REFER TO VAV BOX SCHEDULE ON SHEET M0.2 FOR ADDITIONAL INFORMATION
- B) REFERENCE DIAGRAM ON M0.3 FOR VAV BOX CONTROL.

RS&A

ROBERT SHREVE ARCHITECTS & PLANNERS

MADISON & MAIN BUILDING

801 8TH STREET, SUITE 120

GREELEY, COLORADO 80631

970-346-0151 FAX 970-352-8761

CONSULTING ENGINEERS, INC.

G2

1039 MAIN STREET

WINDSOR, CO 80550

719.586.1000

G2CE.COM

G2CE#2019086

COLORADO REGISTERED PROFESSIONAL ENGINEER

#38512

01/24/2020

DATES

REVISIONS

DESIGNED G2

DRAWN

CHECKED

APPROVED G2

DATE

DRAWING NO. M1.0

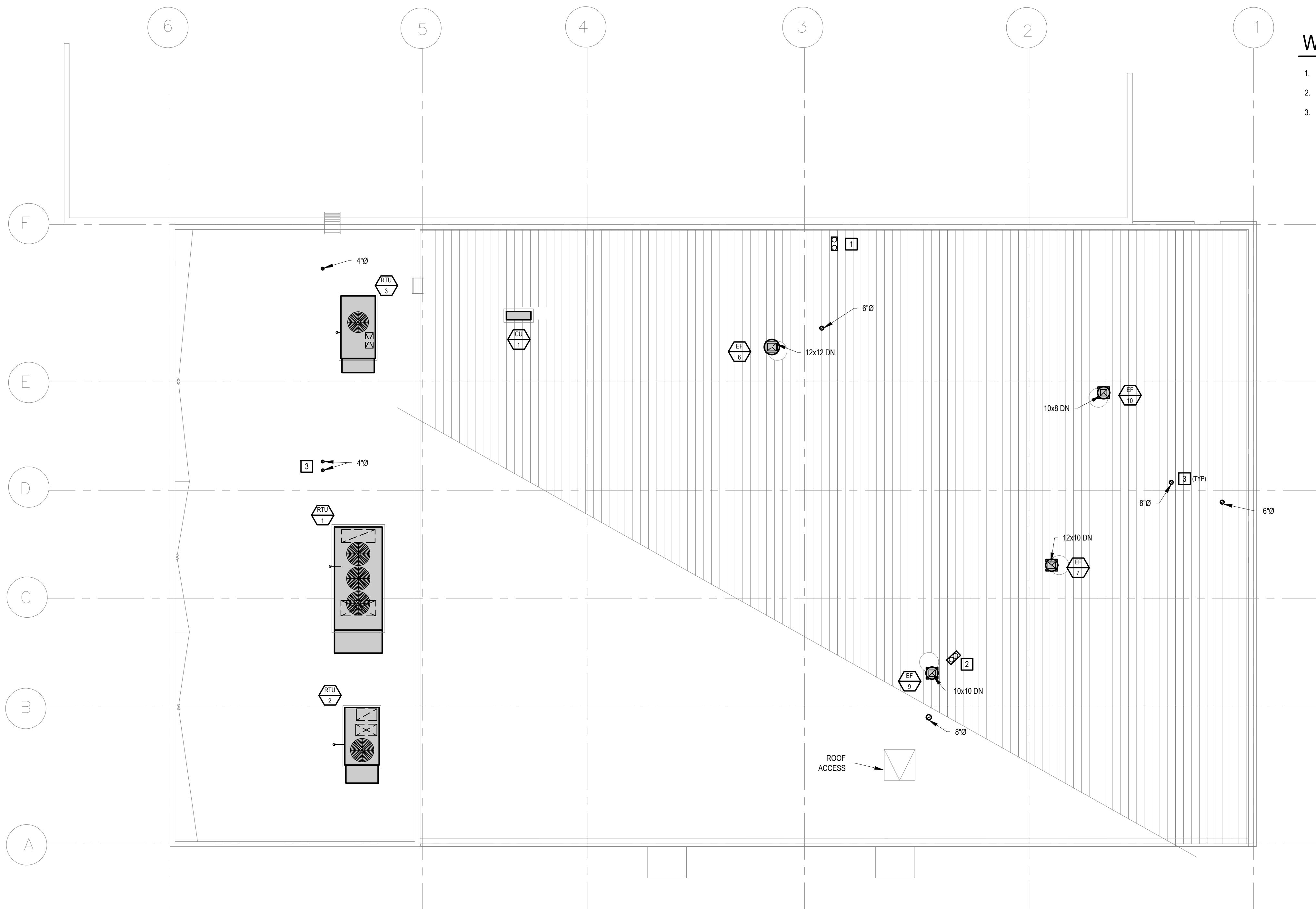
WELD COUNTY

CRIME LAB STORAGE BUILDING

2329 115TH AVE

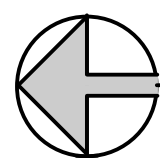
GREELEY, CO.

MAIN FLOOR PLAN - MECHANICAL



WORK NOTES: #

1. DOMESTIC WATER HEATER CONCENTRIC VENT TERMINATION.
2. UNIT HEATER CONCENTRIC VENT TERMINATION.
3. PROVIDE GOOSENECK TERMINATION, TYPICAL.



ROOF PLAN - MECHANICAL

SCALE: 1/8" = 1'-0"

WELD COUNTY
CRIME LAB STORAGE BUILDING
2329 115TH AVE
GREELEY, CO.
ROOF PLAN - MECHANICAL

DATE
DRAWING NO:
M2.0

DESIGNED G2
DRAWN
CHECKED
APPROVED G2

REVISIONS	DATES



CONSULTING
ENGINEERS,
INC.
1039 MAIN STREET
WINSTON, CO 80550
TEL: 970-352-8761
G2CE.COM
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RSAP
ROBERT SHREVE ARCHITECTS & PLANNERS
MADISON & MAIN BUILDING
801 8TH STREET, SUITE 120
GREELEY, COLORADO 80631
970-346-0151 FAX 970-352-8761